

Turtle

Eg1.

```
import turtle as t
t.pencolor("red")
t.pensize(3)
for i in range(50):
    t.forward(i * 5)
    t.right(144)
```



Eg2.

```
import turtle
turtle.hideturtle()
turtle.pen(pencolor="green", pensize=3)
turtle.begin_fill()
for i in range(5):
    turtle.forward(150)
    turtle.right(144)
    turtle.fillcolor("pink")
turtle.end_fill()
```



Eg3.

Write a Python program to print out the following Figure. You can use `style = ("Arial", 15, "italic")` and `turtle.write ("Windows 10 LOGO !", font=style, align="center")`. You can also use `penup ( )`, `pendown ( )`, `goto (90,-200)` for starting the print out text and also use `goto (100,-150)` for displaying yellow rectangle box, `goto (100,-40)` for displaying green rectangle box, `goto (-10,-40)` for displaying red rectangle box and `goto (-10,-150)` for displaying blue rectangle box. Moreover, you can also use `color`, `begin_fill`, `end_fill`, `forward`, `left`, `hideturtle` and `done` function. Beside, define `square( )` function for rectangle box. Otherwise, you can use your own idea for creation.

```
import turtle

turtle.bgcolor('white')

turtle.colormode(255)

def square():

    for i in range (4):

        turtle.forward(100)

        turtle.right(90)

#Display Text

turtle.penup()

turtle.goto(90,-200)

turtle.color("black")

style = ("Arial", 15, "italic")

turtle.write("Windows 10 LOGO!", font=style, align="center")

#yellow box

turtle.left(90)

turtle.penup()

turtle.goto(100,-150)

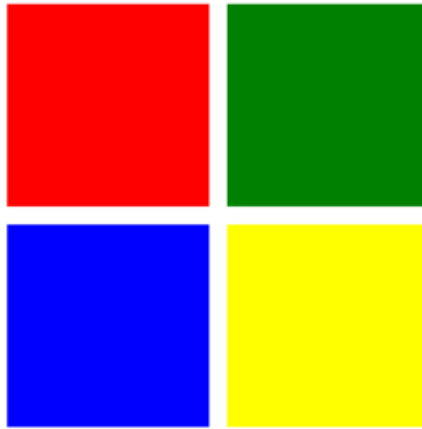
turtle.pendown()

turtle.color('yellow')

turtle.begin_fill()

square()
```

```
turtle.end_fill()
#green box
turtle.penup()
turtle.goto(100,-40)
turtle.pendown()
turtle.color('green')
turtle.begin_fill()
square()
turtle.end_fill()
#red box
turtle.penup()
turtle.goto(-10,-40)
turtle.pendown()
turtle.color('red')
turtle.begin_fill()
square()
turtle.end_fill()
#blue box
turtle.penup()
turtle.goto(-10,-150)
turtle.pendown()
turtle.color('blue')
turtle.begin_fill()
square()
turtle.end_fill()
turtle.hideturtle()
turtle.done()
```



Windows 10 LOGO!

Eg4. Write a program to draw the following figure:

```
import turtle as t
t.pensize(10)
t.fillcolor('yellow')
t.hideturtle()
t.bgcolor('grey')
radius=180
t.penup()
t.setposition(0,-radius)
t.setheading(0)
t.pendown()
t.begin_fill()
t.circle(radius)
t.end_fill()
mouth_radius=radius*0.6
mouth_angle=70
t.penup()
t.setposition(0,-mouth_radius)
```

```
t.setheading(0)
t.pendown()
t.circle(mouth_radius,mouth_angle)
t.penup()
t.setposition(0,-mouth_radius)
t.setheading(0)
t.pendown()
t.circle(mouth_radius,mouth_angle)
t.penup()
t.setposition(0,-mouth_radius)
t.setheading(0)
t.pendown()
t.circle(mouth_radius,-mouth_angle)
x=50
y=50
eye_size=60
t.penup()
t.setposition(x,y)
t.pendown()
t.dot(eye_size)
t.penup()
t.setposition(-x,y)
t.dot(eye_size)
t.done()
```



Eg5. Write a program to draw the following figure:

```
import turtle
screen=turtle.Screen()
trtl=turtle.Turtle()
screen.setup(620,620)
screen.bgcolor('white')
trtl.pensize(4)
trtl.shape('turtle')
trtl.penup()
trtl.pencolor('red')
m=0
for i in range(12):
    m=m+1
    trtl.penup()
    trtl.setheading(-30*i+60)
    trtl.forward(150)
    trtl.pendown()
    trtl.forward(25)
    trtl.penup()
    trtl.forward(20)
    trtl.write(str(m),align="center",font=("Arial", 12, "normal"))
if m==12:
```

```
m=0  
trtl.home()  
trtl.home()  
trtl.setpos(0,-250)  
trtl.pendown()  
trtl.pensize(10)  
trtl.pencolor('black')  
trtl.circle(250)  
trtl.penup()  
trtl.hideturtle()
```

